Evolutive Dataset for Social Housing Design Projects through Artificial Intelligence

From pixel to BIM through deep learning

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Establishing an evolutive dataset for architectural rationalization of social housing is technically achievable through artificial intelligence based on deep learning (DL). However, concerning the sensitive quality of social housing, the application of such technology needs to preserve the human factor and relate ethically to architectural design. A reference on this subject is historic Portuguese research from the 1960s and the 1970s. By then, pioneering research at the National Laboratory of Civil Engineering (LNEC), based in Lisbon, explored early computing methods to aid the design process by considering deontological concerns. The authors studied these works to refactor those goals and concerns of technological application and sociological interaction with current digital technologies. When digitizing their processes of creating architectural design instruments for social housing a problem emerged with parsing a dataset of floor plans and using it to generate building information models. Thus, a DL process was explored to achieve an evolutive dataset in the most automated way at the architectural level. The paper presents the implementation of a DL process that recognizes floor plans of social housing and consequently enables the development of an instrument for direct architectural rationalization.

Keywords: *Artificial Intelligence, Machine Learning, Deep Learning, BIM, Social Housing, Evolutive Dataset.*

INTRODUCTION

Social housing design is an evolving process, especially in its way of relating to the inhabitants. It becomes a problem for the designer to attend to multiple families simultaneously and personalize their demands efficiently without technologies able to process the amount of data and provide helpful design instruments. It becomes more problematic when there is the comprehension that housing and family sociologies often change over time and the designer should correspond to that and act ethically in the design process.

From the sociological side, there is specific research from the 1960s and 1970s about evolutive housing design in Portugal that referenced deontology as an essential component of the process, through the original studies of Paul Henry Chombart de Lauwe (1959) that unfolds in contemporary theoreticians (Léger and Decup-Pannier, 2005; Sabater, 2007; Sabater, 2009; Sabater and Maldonado, 2009; Léger, 2013) and in

architectural practices such as Lacaton & Vassal, Álvaro Siza Vieira, and Alejandro Aravena. The authors had direct access to some of the documents (Portas and Costa, 1966; Pereira, Monteiro, and Portas, 1972) of this research from the National Laboratory of Civil Engineering (LNEC) based in Lisbon and digitized its original information on contemporary sociological standards.

It is important to emphasize that this Portuguese laboratory research was one of the foundations of the Local Ambulatory Support Service (SAAL), one of the primary references of social housing programs in Europe, from 1974 to 1976, and that it was related to a computational process that considered artificial intelligence (AI) already as part of the design methods. Subsequently, this research influenced local expertise in the AI field concerned with ethical standards of the technology (Pereira and Lopes, 2020), providing resourceful means for a humanistic approach.

From the technological side, many studies about the generation of 3D models from 2D floor plan images have been presented in recent years (Dodge, Xu, and Stenger, 2017; Liu et al, 2017; Gerstweiler et al, 2018; Jang, Yu, and Yang, 2020; Dong et al, 2021; Lv et al, 2021; Mishra et al, 2021), offering a view of how 3D modeling, or repetitive design work, could develop shortly. Previous research approached the problem through computer vision and machine learning (ML) procedures and one of them applied ensemble methods (Park and Kim, 2021) understanding the solution for this problem accomplished by the combination of many procedures enchained, which was the benchmark for this research.

The application of deep learning (DL) for using pixel information of floor plan images for digital modeling tackles the optimization of conventional methods of architectural design, for almost a decade. However, remains the alignment with the original LNEC research with the use of AI to assist the design process, instead of choosing other techniques, like shape grammar or space syntax, for example. AI assisting Grasshopper algorithmic modeling that has been studied extensively in the past two decades, and applied to many fields, such as design, architecture, engineering, robotics, and automation.

In this context, an ensemble method was developed for the digitization of the LNEC research using a two-step DL process for parsing the image dataset, alongside the interoperability of Grasshopper (GH) for gathering sociological data and building information modeling (BIM) directly from the floor plan image input. First, it performs an object detection process with the TensorFlow 2 Object Detection API for wall corners. Second, it uses an image-to-image translation process with a TensorFlow 2 Pix2Pix algorithm for wall segmentation of floor plan images. The outputs of these steps are used in the GH algorithmic modeling for the generation of building information models.

This method aids in optimizing the manual work for the LNEC's research digitization process and allows its content to become an evolutive dataset to be updated over time and complemented with more social housing references. The presented dataset is published as an annex of this paper (Lacroix, 2023) and serves as an example. Nonetheless, these methodological procedures could process other floor plan datasets. These results are aligned with the original LNEC research goals of technological instrumentalization and sociological interaction to help architects and designers tackle the difficulties of the social housing process.

METHODOLOGICAL PROCEDURES

This work is part of ongoing research and this paper is a continuation of two previous publications (Lacroix, Lopes, and Sousa, 2021; Lacroix et al, 2022) improving some procedures and adding a new part about training the DL models with custom datasets. Before training the DL models, a previous update of the 1960s floor plan dataset was required, considering current sociological needs and contemporary technical drawing standards. In this sense, the first new procedure was the updating

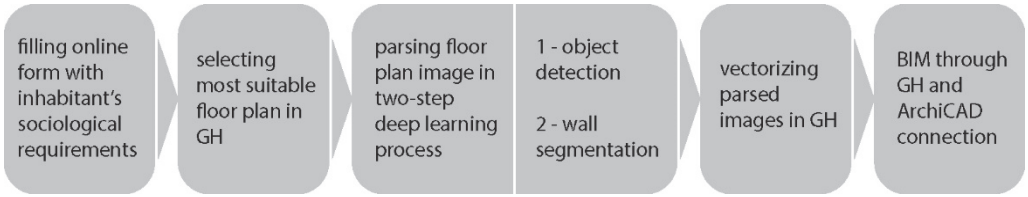

Figure 1
Ensemble method for digitization of the LNEC research.

process which was the first curation regarding many social categories included.

Overview of the ensemble method for the digitization of the LNEC research

The ensemble method (Figure 1) starts with selecting the ideal floor plan accordingly to the hypothetical inhabitant's social needs employing a questionnaire that was converted into an online form for public interaction. The answers are gathered in a spreadsheet that is the input for a GH algorithmic model that finds the most suitable floor plan amongst a dataset of social housing projects, resulting in an image.

Then, the DL method parses the floor plan image resulting in the inputs for a second GH algorithm that enables the BIM of architectural elements, like walls, doors, windows, furniture, etc. The algorithm vectorizes the pixel-based information to generate the primitive geometries for the subsequent BIM.

The social housing project number three from the original LNEC research dataset was selected to illustrate the method in this paper. This project refers to a design from Carlo Aymonino for social housing in Brindisi (Figures 3, 4, 5), Italy, from 1956 to 1963.

Sociological update of the LNEC research social housing floor plan dataset

This stage was crucial, where the curation of the dataset began, and it is a challenge for the designer or architect who is unacquainted with sociological theories, once it requires previous knowledge considered in this original research. Nuno Portas (2004) created a framework for a sociological approach to social housing design based on phenomenology and housing sociology mainly influenced by Chombart de Lauwe. The French sociologist brought the concept of deontology to introduce ethics in the social housing attendance field in the post-war period, and Portas systematized it to comply with design procedures.

In a broad interpretation, deontology is the deconstruction of the essence of the being, in the case of this study the house itself, and its philosophical origins (Bentham, 1834) comprehended the introduction of ethical standards for regulating the professions, a key concept for understanding, for example, the politically correct morality of contemporary society. Besides influencing a whole generation of architects who worked with social housing, this concept is related to creating public policies and legislation for the current practice, something that developed with these kinds of research as its foundations. So, many aspects found in housing references nowadays were not considered in the original LNEC research, and the need to consider a broader range of deontological prerequisites was essential for the curation process.

At least one housing reference was considered by social categories, such as feminism, accessibility, sustainability, co-habitability (Figure 2a), elderly people, LGBTQIAP+ gender equality, African culture, indigenous people, Turkish culture, Brazilian culture (Figure 2b), Chilean culture, and Mexican culture. Some of the examples were not from a recognizable architectural brand but were selected to complement the original dataset focused on European production.

In addition, the selection considered the built works from some contemporary architectural firms with exceptional examples of social housing.

a b c d

Figure 2
a. Cohousing "Un+Un", designed by Catherine Lauvergeat and Thierry Nàberes;
b. Social housing in Brasilia, Brazil, by Nauro Esteves;
c. Neppert Gardens social housing in Mulhouse, France, by Anne Lacaton & Jean-Philippe Vassal;
d. Monterrey social housing in Monterrey, Mexico, by Elemental.

Lacaton & Vassal (Figure 2c), Álvaro Siza Vieira, and Alejandro Aravena (Figure 2d) are references that relate somehow to the evolutive housing system, as by direct influence or by innovating its methods. Zaha Hadid, for bringing radical aesthetics to social housing in Vienna. David Adjaye, for an insertion of African culture in the Bronx.

The architects bring their ideas but also improve the established sociological vocabulary, so the dataset update works at this level but they also help in bringing new kinds of technical drawing representation, the CAD systems, for example, that were embryonary in the 1960s. Therefore, the selected floor plans also contribute to diversifying the drawing patterns, consequently helping with the ML processes.

This affected the whole ensemble method, and both the sociological survey and the social housing selection algorithm were adapted. The sociological update is what makes the dataset evolutive, implementing the original floor plan collection with contemporary standards. Also affects the technological aspect, by diversifying the patterns for image analysis, and ultimately enlarging the dataset with more visual options for the DL steps.

Training deep learning models with custom datasets

There are many working Python notebooks to employ ML procedures available online. In the last decade, these processes became more accessible and many studies have used these techniques for multiple customized purposes.

The authors considered the machine as a designer to help in the BIM stages of social housing design. The AI was programmed to recognize patterns in the floor plan image and to label them with geometric information that will help the parsing from pixel to BIM; it also segments the wall representations of the images to help select the correct lines for wall BIM in GH. Once this two-step DL process is trained, the dataset can be constantly updated with new sociological parameters and new social housing floor plans. Both steps used Google Colab as an environment and Python's TensorFlow 2 as a package library.

The first step employs the Object Detection API with a Faster R-CNN ResNet 101 model, for 1024 by 1024 pixels images, i. e. a region-based convolutional neural network, which labels the floor plan patterns of wall corners with bounding boxes (Figure 3a), providing pixel-based geometrical references for BIM of walls in GH. Thirty floor plans were selected from the LNEC research and thirty floor plans from the sociological update. Sequentially, they were labeled with LabelImg, a Python-based software that provides the dataset metadata through XML. According to the online tutorial (TechZizou, 2021) that shared a Colab notebook with the complete algorithm, the training was fixed for fifty thousand steps of learning. After approximately twenty thousand steps, the training stabilized and stopped in around four hours. This step depends on premium GPU from Google Colab, so it is a matter of resources.

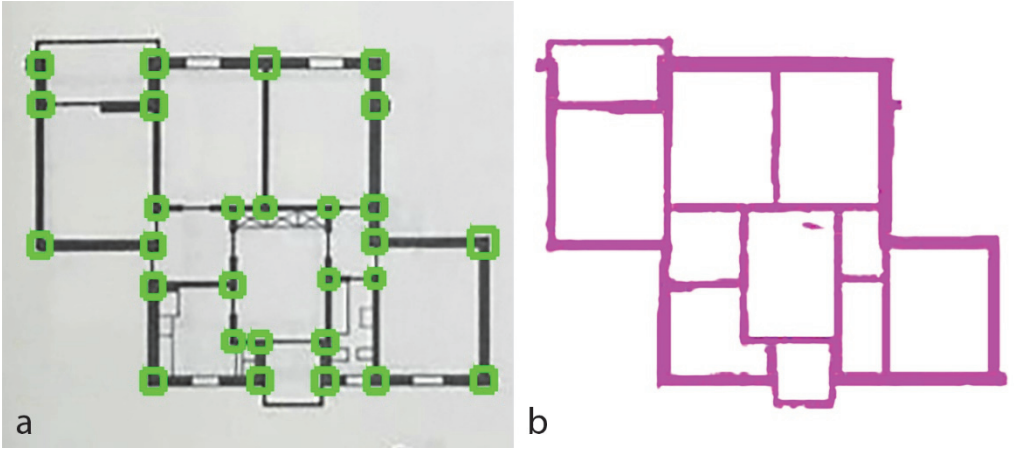

Figure 3
a. Inference from the wall corner object detection trained model;
b. Inference from the wall segmentation pix2pix trained model.

Labeling the dataset (Figure 4) is a procedure that needs particular focus. The dataset was established for 1024 by 1024 pixels images, considering HD format. Then, in LabelImg, the idea was to label the drawing patterns already considering a geometrical arrangement, placing the bounding boxes in the center of the patterns, creating alignments, and maintaining the inherent orthogonalities of the architectural technical drawing. This helps in the precision of the pixel-based process, which can be challenging in some floor plans.

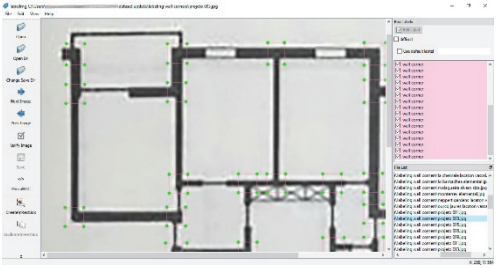

Figure 4
Labeling the dataset in LabelImg software.

The image results from the inference of the object detection trained model are the first inputs for the BIM process in GH. The annotated wall corners in the floor plan image are converted into center points of the bounding boxes and connected through proximity, generating the lines for wall modeling. The proximity component in GH connects all the points; therefore, it needs a second input to help select the lines corresponding to the walls of the floor plan.

The second step employs a Pix2Pix algorithm originally developed by Brownlee (2019) for training a generative adversarial network (GAN) and adapted to segment the walls (Figure 3b) and create a single area of magenta pixels that works as a mask for selecting the lines to the BIM of the walls in the GH algorithmic model interconnected with ArchiCAD (Figure 5). A selection of forty-four images from the previous step was used, mixing the original LNEC dataset with the updated one. The procedure was to mark in all images the elements that constitute the walls of the house with saturated color, to match this segmentation process with the original floor plan images. The training was fixed for one hundred epochs, but it stabilized in the sixtieth epoch, around one hour.

The idea of this segmentation process is to unify the representation of the walls in a single solid area, without the doors and windows gaps, amongst

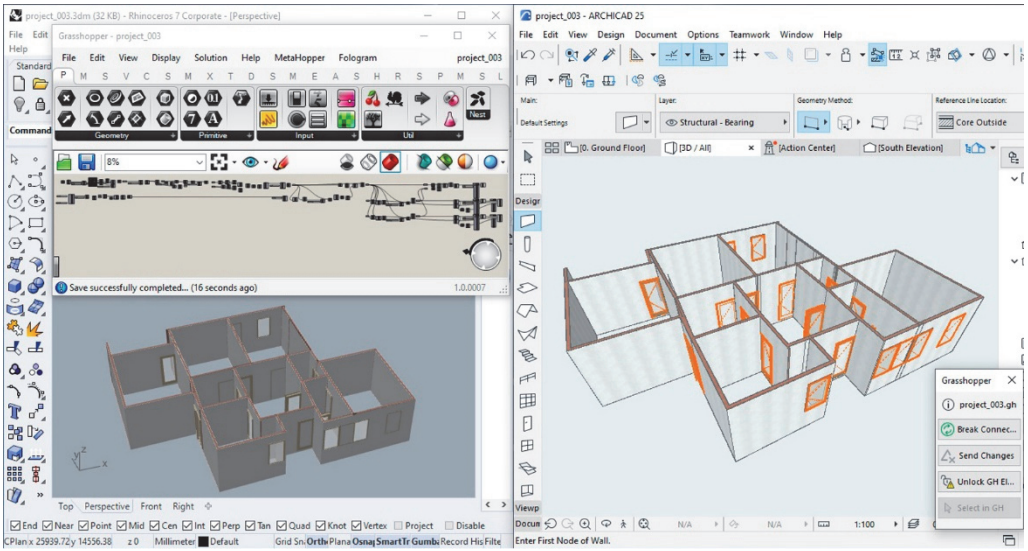

Figure 5
BIM through Grasshopper and ArchiCAD interoperability from pixel-based information directly from the parsed floor plan image.

other patterns such as cobogó, brise soleil, etc. Consequently, it creates the approximate areas of every housing environment separately which helps vectorize them with the GH algorithmic model and use them for a selection process of the lines corresponding to the walls, as a result of the inference of the image-to-image translation trained model.

Evaluation of the deep learning procedures

The employment of AI depends on working with databases to operate its basic functions of letting the machine solve a specific problem independently. If studies such as the LNEC research on housing are to be considered, this difficulty has been around for at least half a century, and in this sense, it cannot be a novelty in the field of architecture. However, the emergent technologies in computer vision, from the last decade, mostly from Google sources, enable different applications and are accessible to the public.

The complexity of the procedures is elevated for most of the students in undergraduate programs because the DL training depends on programming languages that are not provided on a basic level, sometimes not even in graduate programs. In addition, the work with computer vision datasets depends on a lot of manual work before the training procedure, which ended up being the largest and most difficult task for both steps described.

Concerning the first DL training step, the Park and Kim (2021) study helped to understand the functioning of the convolutional neural network (CNN) that analyzes the composition of pixels inside a bounding box, or a label, from its respective center point. In this sense, the areas are labeled to be analyzed and used as sources to identify similar pixel patterns in different images. It is understandable from the mentioned research that the dataset can be smaller in the number of images and may function with lesser training time. The experiment confirmed this on an operative basis, but it needs to be developed in terms of the concision of algorithmic modeling steps and precision of the parsing steps.

Mishra et al (2021) study consolidated the object detection line of research when presented a big data to confirm the robustness of this DL procedure, providing a dataset that enables the modeling of furniture and other elements of the floor plan layout.

For the second DL training step, Chaillou's (2019) research provided comprehensive methods to employ GAN to work on floor plan image recognition, especially in segmentation work. The operation of GAN is easier in comparison to CNN, so it becomes more effective at the architectural level, maybe because of its relation to drawing itself. However, the dataset depends on drawing the segments of walls which can be exhaustive but is achievable through architectural skills. In addition, the GAN needs more amount of different images in the dataset, to cover more visual variability, which can be time-consuming to the data processing. In this sense, the forty-four selected images were rotated 90° three times composing a final dataset of one hundred and seventy-six images.

Both training steps were used to digitize the LNEC research but once the original dataset was accomplished it opened the possibility of working for other images and hence becomes evolutive, other datasets can complement its basis and potentially enable more efficient training, improving its capacities in an evolving way. In addition, it can potentially become interactive and decentralized for public use worldwide, as long as it is updated with information for specific purposes, such as regional climate adaptation, cultural adaptations, technical constraints, and local legislation, for example.

Unfortunately, this paper does not intend to be an entire tutorial for the DL training procedures, but the provision of the digital content available online balances the lack of technical information presented here. The intention is to entwine the main references and share a broader understanding of the process. This is a work in progress and there are many aspects to discuss, but it is feasible for a custom dataset at the architectural level, or the general professional without specialization in the programming field.

DISCUSSION

This research is the final part of a trilogy of experiments that began in 2021, conducted mainly by one architect researcher with knowledge of GH and basic programming skills. In comparison with the research from the 1960s, it was a completely different situation in technological terms. Back then, the organization depended on technicians from multiple fields, mathematics, engineering, architecture, and sociology, demanding sometimes decades of research to accomplish certain tasks.

Nowadays, most of the interfaces present interoperability allowing them to work together, creating unified digital production scripts using ensemble methods. It is not a unique operation, software, or language, but many sequenced parts, optimized for single researchers, requiring less working time. Nonetheless, there is much work for the optimization of this research.

The ensemble method of programming techniques aided by AI procedures is a tool that works on many levels (Dawod and Hanna, 2019; Abhilash and Ahmed, 2023). In this research, GH helped with the interoperability of the process, connecting Microsoft Excel, ArchiCAD, and Rhinoceros, in a digital production script to optimize exhaustive design tasks that become complex when combined in real-life situations, mostly due to the quantity of data to process. The sociological findings are entered using the ensemble method and resulting in the selection of a floor plan image of a social housing project. Then the DL procedure parses this image for direct BIM of the project, enabling digital modeling by sociological information without extensive manual work.

Anyhow, this procedure requires more trained models to BIM other elements such as doors and windows, avoiding the juxtaposition of bounding boxes. Doors and windows were modeled only for illustrative purposes. The idea of using robust object detection (Mishra et al, 2021) with a Mask R-CNN model is a matter of resources, for augmenting the precision of BIM and enabling other elements, such as furniture, to be modeled in an automated way.

One hypothesis for improving the BIM precision is to train a model such as CenterNet HourGlass104 or CenterNet Resnet50 V1 FPN that provides keypoints. The keypoints would deliver geometrical information that would replace an entire algorithmic modeling procedure to find the center point of the bounding boxes.

Another hypothesis is to replace the social housing selecting algorithm in GH with a sociological survey through prompt-to-image VQGAN + CLIP technology, which would enable the creation of new floor plans. However, we argue about the quality of the results without big data, which requires powerful computing capacity. Another challenge is to train the prompt with custom sociological data to implement natural language processing (NLP) with specific metadata, considering that achieving an architectural level of floor plan solution requires description specificities from many fields of knowledge, like psychology, philosophy, ethics, etc., that the updated sociological survey of the LNEC research already provides, at least a framework.

The machine may create a representation of a floor plan with a prompt-to-image AI, but senseless in comparison to the multiple factors that designed the references that the dataset provides, which, in the end, may not be effective for real purposes. We went here as far as we think it is scientifically reliable to help in real practice, and it is open for those entities who work with similar datasets to use these procedures on a large scale, helping them curate the datasets on an ethical basis.

From the ethical standpoint, this evolutive dataset is an open-source non-financial activity, which is expected to interact with those that are related to helping less-favored populations through the difficult process of having their social right implemented in reality. Therefore, the idea of preserving the intellectual properties of these references is to guarantee the efficacy of the solutions, since most of them were built, and tested over time, with an architectural intelligence that the machine is not able to provide unless extensive resources are available, which are unlikely to happen in these social situations.

From a decolonial standpoint, it is necessary to reflect on the importance of a European method for worldwide application. Our opinion is that the work coordinated by Nuno Portas represents an exception that lasted for only a couple of years as a real policy in a politically changing Portugal, in the 1970s. The artist and researcher María García Ruiz (2023) understands the SAAL as a revolutionary architecture, which is linked directly to the extensive LNEC research on housing from the 1960s and 1970s. Moreover, the basis of this research was humanistic, to improve in reality the quality of less favored populations, and rooted in a philosophical grounding that relates already to deontology.

The present research is far from becoming a product, but it has never been closer to becoming operative in more than 50 years, so we believe that this work in progress only depends on the effort of people who care about this kind of approach. The effort is not only technical but depends on the curation process, the selection of parameters, categories, and concepts to be considered, and the development of the sociological survey. It should be a multidisciplinary research, taking the example of the LNEC organization itself.

CONCLUSION

The paper presented a method that enabled the optimization of the digitization of the LNEC research from the 1960s and 1970s and developed it into an evolutive dataset of social housing designs and works updatable over time. The method contributes to providing a framework for the curation process that this evolutive dataset requires, assuring the preservation of architectural intelligence that has been gathered so far regarding social housing.

The method can potentially become a platform for easy BIM supported by floor plan images. It contributes to the research of 3D modeling from 2D floor plan images employing BIM with the deep learning ensemble methodology.

The possible directions of future research are:

- To train an object detection model that provides keypoints to improve the BIM procedure's precision and optimize the algorithmic modeling in GH.
- To use the ShapeDiver add-on for GH to insert the algorithmic model online.
- In addition, to make the inference of the deep learning models available online to be able to provide the inputs for BIM in an integrated system.

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